

EC-Council Certified SOC Analyst (C|SA)

Certified SOC Analyst Certification

This course is focused on:

- Security Operations Center (SOC) Analyst Training
- C|SA Exam Candidates

Course Information:

- Exam Code: 312-39
- Number of Questions: 100
- Test Duration: 3 Hours
- Test Format: Multiple Choice
- Availability: EC-Council Exam Portal (<https://www.eccexam.com>)

Course Overview

The EC-Council Certified SOC Analyst (C|SA) program equips learners with essential skills in security operations, threat intelligence, and incident response. It covers the processes, technologies, and techniques used to detect, investigate, and respond to threats, as well as attack vectors, SIEM deployment (with 350 use cases), and SOC development. Students gain proficiency in Centralized Log Management, incident triage, IoC investigation, and the application of the cyber kill chain. They also learn to create effective reports and leverage AI-enabled tools and platforms to enhance SIEM capabilities, automate threat detection, prioritize alerts, and support threat hunting.

- Build job-ready skills with 50 labs and 120 tools
- Earn a globally recognized, in-demand certification
- Learn flexibly without leaving your current job

Module 01: Security Operations and Management

Learn how a SOC enhances an organization's security management to maintain a strong security posture, focusing on the critical roles of people, technology, and processes in its operations.

- SOC Overview and Capabilities
- SOC Operations and Workflow
- Components of SOC
- SOC Models
- SOC Maturity Models
- SOC Generations
- SOC KPIs and Metrics
- SOC Challenges

Module 02: Understanding Cyber Threats, IoCs, and Attack Methodology

Learn various cyberattacks, their IoCs, and the attack tactics, techniques, and procedures (TTPs) cybercriminals use.

Hands-on Labs:

- Perform SQL injection attack, Cross-Site Scripting (XSS) attack, network scanning attack, DoS attack, and brute force attack to understand their TTPs and IoCs
- Detect and analyze IoCs using Wireshark

Key Topics:

- Cyber Threats and TTPs
- Reconnaissance Attacks
- Man-in-the-Middle Attacks
- Password Attack Techniques
- Malware Attacks
- Advanced Persistent Threat Lifecycle
- Host-Based DoS Attacks
- Ransomware Attacks
- SQL Injection Attacks and XSS Attacks
- Cross-Site Request Forgery (CSRF) Attack
- Session Attacks and Social Engineering Attacks
- Email Attacks and Insider Attacks
- Indicators of Compromise (IoCs)
- Attacker's Hacking Methodology
- MITRE D3FEND Framework
- Diamond Model of Intrusion Analysis

Module 03: Log Management

Learn log management in SIEM, including how logs are generated, stored, centrally collected, normalized, and correlated across systems.

Hands-on Labs:

- Configure, monitor, and analyze various logs
- Collect logs from different devices into a centralized location

Key Topics:

- Incident, Event, and Log Overview
- Log Sources and Log Format
- Local Logging – Windows Event Log, Linux Logs, Mac Logs
- Network Device Logs – Firewall, iptables, Router, IIS, Apache
- Database Logs
- Centralized Logging – Log Collection, Transmission, and Storage
- AI-Powered Script for Log Storage
- Log Normalization and Parsing
- Log Correlation, Analysis, Alerting and Reporting

Module 04: Incident Detection and Triage

Learn SIEM fundamentals, including its capabilities, deployment strategies, use case development, and how it helps SOC analysts detect anomalies, triage alerts, and report incidents.

Hands-on Labs:

- Develop Wazuh use cases to detect and generate alerts for brute-force attempts, ransomware attacks, SQL injection, XSS, Broken Access Control, and more
- Enhance alert triage using Wazuh
- Create dashboards in Wazuh
- Create ELK use cases for monitoring trusted binaries, credential dumping using Mimikatz, and malware activity
- Detect brute-force attack patterns using correlation rules in ManageEngine Log360

Key Topics:

- SIEM Architecture and Its Components
- AI-Enabled SIEM
- Types of SIEM Solutions
- SIEM Deployment and Use Cases

- SIEM Deployment Architecture
- SIEM Use Case Lifecycle
- Application-Level, Insider, and Network-Level Incident Detection Use Cases
- Compliance Use Cases
- SIEM Rules Generation with AI
- Alert Triage with AI
- Dashboards in SOC and SOC Reports

Module 05: Proactive Threat Detection

Learn the importance of threat intelligence and threat hunting for SOC analysts, and how their integration with SIEM helps reduce false positives and enables faster, more accurate alert triage.

Hands-on Labs:

- Integrate IoCs into the Firewall
- Integrate Suricata into the Firewall
- Detect incidents in Windows Server 2025
- Conduct threat hunting using OSINT

Key Topics:

- Cyber Threat Intelligence (CTI)
- Threat Intelligence Lifecycle and Types
- Threat Intelligence Strategy and Sources
- Threat Intelligence Platform (TIP)
- Threat Intelligence-Driven SOC
- Threat Intelligence Use Cases for Enhanced Incident Response
- Enhanced Threat Detection with AI
- Threat Hunting Process and Frameworks
- Threat Hunting with PowerShell Script and AI
- Threat Hunting with YARA
- Threat Hunting Tools

Module 06: Incident Response

Learn the stages of incident response and how the IRT collaborates with SOC to handle and respond to escalated incidents.

Hands-on Labs:

- Generate tickets for incidents
- Contain data loss incidents
- Eradicate SQL injection and XSS incidents
- Perform recovery from data loss incidents
- Create incident reports using OSSIM
- Perform automated threat detection and response using Wazuh
- Detect threats using Opensense Firewall
- Integrate Opensense Firewall with Wazuh

Key Topics:

- Incident Response (IR) and IRT
- SOC and IRT Collaboration
- IR Process – Ticketing, Incident Triage, Notification
- Containment, Eradication, and Recovery
- Network Security Incident Response
- Application Security Incident Response
- Email Security Incident Response
- Insider Threats and Malware Threats Incident Response

- SOC Playbook
- Endpoint Detection and Response (EDR)
- Extended Detection and Response (XDR)
- SOAR and SOAR Playbook

Module 07: Forensic Investigation and Malware Analysis

Learn the importance of forensic investigation and malware analysis in SOC operations to understand attack methods, identify IoCs, and enhance future defenses.

Hands-on Labs:

- Perform forensic investigation of application security incidents: SQL injection attacks
- Perform forensic investigation of a compromised system
- Analyze RAM for suspicious activities using Redline
- Perform static analysis on a suspicious file using PeStudio
- Examine a suspicious file using VirusTotal
- Perform dynamic malware analysis in Windows using Process Hacker

Key Topics:

- Forensics Investigation Methodology and Process
- Forensics Investigation of Network Security Incidents
- Forensics Investigation of Application Security Incidents
- Forensics Investigation of Email Security Incidents
- Forensics Investigation of Insider Incidents
- Malware Analysis – Types and Tools
- Static Malware Analysis
- Dynamic Malware Analysis

Module 08: SOC for Cloud Environments

Learn the SOC processes in cloud environments, covering monitoring, incident detection, automated response, and security in AWS, Azure, and GCP using cloud-native tools.

Hands-on Labs:

- Implement Wazuh

Key Topics:

- Cloud SOC Architecture

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Accreditations & Recognitions

- 100% Compliant with NICE 2.0 Framework – Cyber Defense Analysis (CDA) role under Protect and Defend (PR) category
- Endorsed by the American Council on Education (ACE)
- Recognized by the U.S. Army and DoD
- Certified professionals recognized in 150+ countries across organizations including Accenture, Cisco, IBM, Microsoft, TCS, and Xerox